

WAA | Board Review 2026

Primary Care Optometry — Consolidated Review

Personal study notes | Sources: Dr. Ferrer PCO Complete Notes (MCU 2024) · PCO Board Reviewer 10 Sessions · PrimaryCareOptometry Program Assessment Reviewer

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1. Examination Sequence Overview

Primary care optometry follows one logical sequence. Each step informs the next — none are independent.

Phase	Area	Key Tests
1	Case History	Chief complaint, LOFDAR, POMR
2	Refraction	VA, retinoscopy, subjective refraction
3	Binocular / Accommodative Status	NPC, phorias, vergences, PRA/NRA
4	Ocular Health	External exam, slit lamp, ophthalmoscopy, IOP

Clinical Protocol: Hand hygiene before and after every patient. Disinfect instruments between patients using 70% isopropyl alcohol or 0.5% sodium hypochlorite. IPAO/COVID-19 guidelines require PPE, health screening, and physical distancing.

2. Case History

Definition and Purpose

Case history is the patient's account of their visual and ocular problems. It builds rapport, organizes exam direction, and generates initial impressions before any testing begins.

4 POMR Components (Weed, 1968–1969) – "CAMP"

Letter	Component	Covers
C	Chief / Problem List	Complaints the patient experiences or observes
A	Assessment / Initial Treatment Plan	Provisional diagnosis and management plan
M	Medical Records (Defined Data Base)	Patient profile, history, medications
P	Progress Notes	Follow-up records and response to treatment

Essential Elements of Case History

1. Patient profile (name, age, address, gender, occupation, education)
2. Chief complaint – primary reason for visit
3. Visual and ocular history (present and past)
4. General health history
5. Medication usage and allergies
6. Family ocular and medical histories
7. Occupational and recreational visual requirements

Clinical significance of patient profile fields:

Field	Why It Matters
Age	Influences diagnosis, prescription, treatment (e.g., presbyopia at 40+)
Address	Socio-economic status, disease prevalence in area
Gender	Disease predilection differs by sex
Occupation / Hobby	Determines lens type, amount of correction, disease predilection
Education	Guides communication style and IQ-appropriate explanations

Chief Complaint — Asthenopia

Asthenopia: uncomfortable, painful, irritable vision from any form of functional eyestrain.

Sources of ocular complaints:

Source	Cause
Accommodative	Uncorrected refractive error
Motor	Muscular deficiency
Integrative	Combination of refractive error AND muscular deficiency

Memory hook: **AMI** — Accommodative, Motor, Integrative.

LOFDAR — Describing Any Complaint

Use LOFDAR to fully characterize any complaint during history:

- **L** — Location (laterality: right, left, both; where on the eye)
- **O** — Onset (sudden or gradual)
- **F** — Frequency (how often)
- **D** — Duration (how long each episode lasts)
- **A** — Alleviating / Aggravating factors
- **R** — Related / Associated signs and symptoms

Headache Location and Refractive Cause

Location	Most Likely Cause
Frontal	Hyperopia
Occipital	Presbyopia or convergence insufficiency
Temporal	Astigmatism
Parietal	Tension headache

Common Complaints and Probable Causes

Complaint	Most Likely Cause	Important Note
Blurred vision	Uncorrected refractive error, cataract, diabetic retinopathy	Ask: when does it occur?
Headache	Uncorrected refractive error, stress, migraine	Location and triggers matter
Diplopia	Weak EOM, high astigmatism, uncorrected refractive error	Monocular vs. binocular
Photophobia	Albinotic / lightly pigmented eyes, retinal degeneration	Also: uveitis, corneal abrasion
Itching	Allergy, dry eye	Rule out seasonal pattern
Floaters	Usually physiologic; vitreous hemorrhage, retinal detachment	Sudden onset = urgent referral

Systemic Effects on Refractive State

Blood Sugar	Lens Effect	Refractive Change
High (hyperglycemia)	Lens swells	MORE MYOPIC / less hyperopic
Low (hypoglycemia)	Lens shrinks	MORE HYPEROPIC / less myopic

Memory rule: **HIGH sugar = INFLAMED lens = MORE MYOPIC** (focal point shifts anterior).

Communication Skills — 8 Key Behaviors

1. Greet with a smile; introduce yourself before asking the patient's name
 2. Use open-ended questions; move from general to specific
 3. Active listening — attend to verbal and non-verbal cues
 4. Silence — allow time to respond; avoid rushing
 5. Empathy and warmth — treat the patient as a person, not a clinical problem
 6. Sensitivity to psychosocial and personal issues
 7. Clear and concise questions in layman's terms
 8. Avoid repetition and redundancy (especially with children and elderly)
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3. Visual Acuity

Definition

Visual acuity (VA) is the clarity or sharpness of vision — the ability to discriminate the contour and detail of objects. From the Latin *acuitas* (sharpness). Dr. Herman Snellen (1834–1908), Dutch ophthalmologist, created the standard test chart.

Forms of Visual Acuity — Lowest to Highest

Form	Definition	Example
Minimum Visible (Detection)	Recognizing a figure against its background	Detecting a dot on a blank field
Minimum Separable (Resolution)	Distinguishing two lines as separate; normally 5' of arc	Tumbling E, Landolt C — Snellen standard
Minimum Cognoscible / Legible (Recognition)	Threshold for identifying letters	Snellen chart, Sloan letters
Minimum Separable Angle	Smallest angle subtended at nodal point for two points seen as separate	Grating acuity

Memory hook: **VSCS** — Visible, Separable, Cognoscible, Separable Angle.

Board note: Snellen acuity is most often classified as minimum separable. Some sources define it as minimum cognoscible.

VA Terminology

- **Unaided VA** — vision without any correction; baseline data
- **Habitual VA** — vision with the patient's current correction
- **Optimal VA** — best corrected VA after full refraction
- **Pinhole VA** — acuity through a pinhole disc; bypasses refractive error

Snellen Chart Design Standards

Parameter	Standard
Background / contrast	Black letters on white; 80% contrast
Luminance at patient's eye	20 foot-candles (20 fc)
Test distance	20 feet (6 metres)
Letter stroke width	1/5 of the 5-minute square
Letter spacing	1/2 the width of the square
Row separation	1/2 the size of the square
Minimum letters per row (20/50 and smaller)	5 letters
Scoring stop point	Row where more than half the letters are missed
Easiest letter	L
Hardest letter	B

Snellen chart construction formula: $\tan 5' = X/\text{distance}$. The basis is a square of 5' of arc in each dimension with letters formed in squares each subtending 1' of arc.

Types of Snellen Charts

- Standard Snellen (letters)
- ETDRS (Early Treatment Diabetic Retinopathy Study) — gold standard for research
- Landolt C — resolution standard
- E-test (tumbling E) — for preliterate or non-verbal patients

- Bailey-Lovie — logarithmic spacing

Snellen Notation Systems

System	Format	Example
Imperial (English)	Feet	20/20
Metric	Metres	6/6
Decimal (International)	Decimal	1.0
LogMAR	Log of minimum angle of resolution	0.0

Conversion: 20 feet = 6 metres. Decimal = numerator ÷ denominator. 20/50 = 0.4; 20/200 = 0.1.

Expected VA for Uncorrected Refractive Error (Borish)

Snellen VA	Uncorrected Sphere Error	Uncorrected Cylinder Error
20/20	< 0.25 D	< 0.25 D
20/30	0.50 D	1.00 D
20/40	0.75 D	1.50 D
20/60	1.00 D	2.00 D
20/80	1.50 D	3.00 D
20/120	2.00 D	4.00 D
20/200	2.00–3.00 D	5.00 D

Procedure: Distance VA

1. Patient at 20 feet (6 m) from chart under 20 fc illumination
2. Occlude the LEFT eye; test OD first
3. Patient reads from largest letter downward
4. Record the last row where more than half the letters are correct
5. Note any missed letters: 20/30-1 (missed one), 20/30+2 (read two extra)
6. Repeat for OS; then test OU last

Subnormal Vision Techniques — In Order

Step	Technique	Procedure	Recording
1	Move patient closer	Move until they can read the largest letter; min 0.5 m	20/200 at ____ ft
2	Counting Fingers (CF)	Expose fingers at ~1 ft	OD CF at 3 ft
3	Hand Motion (HM)	Ask if patient sees hand movement	OD HM at 4 ft
4	Light Projection (LP)	Penlight in 4 quadrants at 20 inches	OD LP
5	Light Perception (LP)	Direct penlight — can patient detect light? Monocular	OD LP / NLP

Factors Affecting VA

Environmental: Intensity of illumination, spectral nature of light, surrounding field intensity, region of retina stimulated, lateral illumination and glare, distance of test target.

Ocular: Lid pressure, tear film, pupil size (optimal 2–4 mm in light), presence of ametropia, retinal mosaic, clarity of optical media.

4. Preliminary Examination

Purpose

Assesses visual abilities, ocular health, and related systemic health status — provides a complete initial view of the patient before refractive testing begins.

4.1 Interpupillary Distance (IPD / PD)

Distance between the centres of the entrance pupils. Also called Interocular Distance (IOD) or Pupillary Distance (PD). Recorded in millimetres with distance PD first (e.g., 64/61).

Clinical significance:

- Places optical centres of spectacle lenses accurately to avoid unwanted prismatic effect
- Errors in PD produce induced prism and cause eyestrain
- Distance PD is always larger than near PD

Typical values:

- Distance PD: 60–68 mm
- Near PD: 3 mm less than distance PD

Methods:

1. **Anatomical Method** — using a millimetre ruler
 - Close your RIGHT eye; patient fixates on your open LEFT eye
 - Align zero to the nasal border of patient's RIGHT pupil
 - Read at temporal border of patient's LEFT pupil
 - Close your LEFT eye; patient fixates your RIGHT eye (near PD)
2. **Catoptric Method** — using the corneal reflection of a penlight

Near PD Formula:

$$\text{NPD} = (\text{DPD} \times \text{WD}) / (\text{WD} + 27)$$

The constant **27** = 13.5 mm (behind cornea) + 13.5 mm (to spectacle plane).

Proportional formula:

$$\text{NPD} = (\text{DPD} / \text{WD}) \times (\text{WD} - 27)$$

Indications for monocular PD measurement:

- Unilateral strabismus
 - Facial asymmetry
 - High refractive error (horizontal tolerance: 2 mm; vertical tolerance: 1.2 mm)
 - Multifocal dispensing
-

4.2 Near Point of Convergence (NPC)

The point where visual lines intersect at maximum convergence — measures the maximum convergence ability.

Instrument: RAF Rule (Royal Air Force) — primary instrument; target placed at 45°. Also: penlight, pencil, Wolff ball.

Normal values (Scheiman and Wick, 2008):

Target Type	Break	Recovery
Accommodative target	5 cm ± 2.5	7 cm ± 4
Penlight (non-accommodative)	7 cm ± 4	10 cm ± 5

Amplitude of Convergence Formula:

$$\text{Amplitude of Convergence} = (\text{NPD} \times 10) / (\text{NPC} + 3)$$

Example: NPD = 60, NPC = 7 → $(60 \times 10) / (7 + 3) = 600 / 10 = 60$ prism dioptres

4.3 Pupil Size Evaluation

Condition	Typical Size
Room light	4.0 mm
Near darkness	6.5 mm
Direct light	3.0 mm

Anisocoria: Difference in pupil size between the two eyes.

- Physiologic anisocoria: <0.5 mm difference, normal in ~20% of population
- Pathologic: >1 mm, associated with CN III palsy, Horner's syndrome, Adie's tonic pupil

4.4 Fixation Test / Corneal Reflex Test (Hirschberg Method)

Determines the presence of strabismus. Light held at 40 cm. The corneal reflex in the normal eye is slightly nasal (0.5 mm nasal to pupil center) because the visual axis and optical axis are not coincident.

Interpretation of reflex position:

Reflex Position	Diagnosis
Slightly nasal	No deviation (normal)
Extremely nasal	Exodeviation
Temporal	Esodeviation

Hirschberg quantitative equivalents:

Corneal Reflex at Pupil Edge	Prism Dioptres	Degrees
Pupil edge	~10Δ	5°
Mid-iris	~30Δ	15°
Limbus	~60Δ	30°
Beyond limbus	~90Δ	45°

Angle kappa: The angle between the visual axis and the pupillary axis. A positive angle kappa gives the appearance of exotropia (pseudosstrabismus).

4.5 Version Test / Ocular Motility

Tests conjugate eye movements using the 6 extraocular muscles. Uses H-pattern (6 diagnostic gaze positions plus primary position).

Governing laws:

- **Sherrington's Law of Reciprocal Innervation:** When an agonist contracts, the antagonist relaxes
 - **Hering's Law of Equal Innervation:** Both eyes receive equal simultaneous innervation to yoke muscles
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4.6 Pupillary Reflexes

Direct Light Reflex (DLR): Shining light into an eye causes that eye's pupil to constrict. Afferent: CN II; efferent: CN III.

Indirect / Consensual Light Reflex (ILR): Shining light into one eye causes the opposite eye's pupil to constrict. Tests the consensual pathway via the optic chiasm.

Near / Accommodative Reflex (PNR): Pupil constricts with near fixation. Part of the near triad: accommodation + convergence + miosis.

Sample recording: DLR = 3+/3+ | ILR = 2+/2+ | PNR = 2+/2+ | PERRL (or PERLA)

Grading:

- 4+ = brisk, very rapid constriction
 - 3+ = normal, moderately rapid
 - 2+ = sluggish
 - 1+ = barely detectable
 - 0 = absent
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4.7 Swinging Flashlight Test (Marcus Gunn Pupil / RAPD)

Detects unilateral retinal or optic nerve anomalies anterior to the lateral geniculate nucleus. Rapid, inexpensive, reliable.

Procedure: Swing light alternately between eyes in a darkened room, allowing 1–2 seconds per eye.

Positive test (Marcus Gunn Pupil / RAPD): Paradoxical DILATION of the impaired eye when the light swings to it. This indicates an afferent pupillary defect — the affected eye's optic nerve transmits a weaker signal, so when the light moves to it, both pupils dilate relative to their state when the good eye was illuminated.

4.8 Color Vision Testing

Types of color vision deficiency:

Type	Description
Monochromacy	Only one type of cone (or rod monochromacy – no cones)
Dichromacy	Two cone types functional; one absent
Anomalous Trichromacy	All three cone types present but one shifted in spectral sensitivity

Dichromacy types:

Type	Cone Absent	Color Axis
Protanopia	Red (L-cone)	Vertical
Deuteranopia	Green (M-cone)	Slant
Tritanopia	Blue (S-cone)	Horizontal

Anomalous trichromacy: Protanomaly (red weak), Deuteranomaly (green weak, most common overall), Tritanomaly (blue weak).

Acquired color vision deficiencies:

Lesion Location	Deficiency Type
Choroid	Yellow-blue (tritan)
Retina	Yellow-blue OR red-green
Optic nerve	Red-green

Ishihara Pseudoisochromatic Plates (1917)

Tests red-green color deficiencies. Standard version: 38 plates.

Plates	Type	Normal	Deficient
Plate 1	Demonstration	Number	Same number (everyone sees this)
2–7	Transformation	Correct number	A different number
8–13	Vanishing	Number present	Cannot see the number
14–15	Hidden design	No number	A number appears (diagnostic)
16–17	Diagnostic severity	Varies	Determines severity
18–24	Tracing	Tracing path	For illiterate / children

Procedure:

- Patient VA must be better than 6/60 (0.1)
- Good illumination (Macbeth Easel Lamp recommended)
- Test monocularly
- Hold chart 75 cm from patient, perpendicular to visual axis
- Allow 3–5 seconds per plate
- Patient may trace with a cotton tip if unable to identify numbers
- **Scoring:** 4 or fewer incorrect = normal; 5 or more incorrect = defective red-green color vision
- Example recording: 17/18 (normal); 10/18 (defective)

Board trap: Ishihara tests ONLY red-green deficiencies. It does NOT test blue (tritan) defects.

Hardy Rand Rittler (HRR) Test

Tests both red-green AND blue-yellow (tritan) deficiencies. More comprehensive than Ishihara.

Farnsworth D-15

- 16 color caps (one fixed reference)
- Tests for color confusion axes
- Failure criterion: **2 or more crossings involving caps at least 4 numbers apart**
- Color axes: Protan = vertical, Deutan = slant, Tritan = horizontal

Farnsworth-Munsell 100 Hue (FM-100)

- 85 color caps arranged in 4 trays
- Gold standard for mild CVD and acquired color deficiencies

- Tests the full color discrimination spectrum
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4.9 Visual Field

The entire area seen when the eye is directed on a single point, including peripheral vision.

Confrontation test: Gross screening method comparing patient's field to examiner's.

Amsler Grid

- Held at **33 cm** from patient; patient wears habitual near correction
- Patient fixates the center dot and observes all 4 corners, line straightness, and missing or blurry squares
- Each small square subtends 1° of arc at 33 cm; full grid covers the central 10° (20° total diameter)

Finding	Meaning
Metamorphopsia	Lines appear wavy or distorted
Macropsia	Squares appear larger than normal
Micropsia	Squares appear smaller than normal
Scotoma	Area of missing squares

4.10 Near Point of Accommodation (NPA) — Amplitude of Accommodation

Push-up technique: Move a near target toward the eye until it first blurs.

$$\text{Amplitude} = 100 / \text{NPA distance (in cm)}$$

Example: NPA at 25 cm $\rightarrow 100/25 = \mathbf{4.00\ D}$

Hofstetter Formulas:

Type	Formula
Minimum	$15 - 0.25 \times \text{age}$
Average	$18.5 - 0.30 \times \text{age}$
Maximum	$25 - 0.40 \times \text{age}$

Quick check at age 40: Minimum = $15 - (0.25 \times 40) = \mathbf{5.00\ D}$

Donders Table (key values):

Age	Expected Amplitude
10	14.0 D
20	10.0 D
30	7.0 D
40	4.5 D
50	2.5 D
60	1.0 D

4.11 Dominant Eye Test

Tests perceptual or motor superiority of one eye.

- **Ring Test:** Patient fixates examiner's nose through a ring formed by their fingers
 - **Hole-in-a-Card Test:** Patient sights through a small hole in a card
 - **Box Test:** Patient aligns two pipe cleaners
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4.12 Amblyopia

Current definition: unilateral or bilateral decrease of visual acuity caused by pattern vision deprivation or abnormal binocular interaction for which no obvious cause can be found by physical examination and which cannot be corrected by optical or surgical means, but is reversible by therapeutic measures.

Criteria: difference of >2 lines between two eyes, OR VA <6/9.

5. External Ocular Health Examination

7 Rules for Recognizing Healthy Eyes

- 1. The cornea should be clear
- 2. The pupil should be black
- 3. The sclera should be white
- 4. The eyelids should open and close properly
- 5. The eyelashes should point away from the eye
- 6. Children's eyes should appear straight
- 7. The vision should be normal

Areas Assessed in External Exam

Facial expressions and symmetry, skin color and texture, hair color and texture, gross eye movements and position, head position (a tilt or turn may indicate EOM compromise), eyelid position.

Corneal Anomalies

Corneal opacity classification:

Type	Description	Effect on Vision
Nebula	Slight haze; seen only by oblique illumination	Seldom obstructs vision
Macula	Moderately dense, circumscribed whitish opacity	May reduce VA
Leukoma	Opaque white spot	Significant VA loss

Keratoconus: Progressive ectatic disorder causing central or paracentral corneal thinning and anterior protrusion. Characteristic Munson's sign (V-shaped indentation of lower lid on downgaze). On retinoscopy: scissors motion.

Conjunctival Anomalies — Conjunctivitis Types

Type	Features
Allergic	Bilateral; itching; papillary reaction; watery/ropy discharge; chemosis
Viral	Watery discharge; follicular reaction; pre-auricular lymphadenopathy
Bacterial	Mucopurulent discharge; papillary reaction; morning crusting
Giant Papillary Conjunctivitis (GPC)	Associated with contact lens wear; giant papillae (>1 mm) on upper tarsal conjunctiva
Ophthalmia Neonatorum	Neonatal conjunctivitis; within 30 days of birth; gonorrheal (day 1–5), chlamydial (days 5–14)

Eyelid Anomalies

Condition	Description
Ptosis	Drooping of upper eyelid; can be neurogenic (CN III palsy, Horner's) or myogenic
Hordeolum	Acute staphylococcal infection; internal (meibomian gland) or external (Zeis/Moll gland)
Chalazion	Chronic lipogranuloma of meibomian gland; painless, firm nodule
Blepharitis	Chronic inflammation of eyelid margins; anterior (staphylococcal, seborrheic) or posterior (meibomian gland dysfunction)
Lagophthalmos	Inability to fully close the eyelid; exposes cornea to drying and abrasion
Ectropion	Outward turning of eyelid margin

Entropion	Inward turning of eyelid margin; lashes may contact globe
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Eyelash Anomalies

Condition	Description
Trichiasis	Misdirected lashes growing inward; rubs against cornea
Distichiasis	Accessory row of lashes from meibomian gland orifices
Poliosis	Localized depigmentation of lashes; associated with Vogt-Koyanagi-Harada, sympathetic ophthalmia

Iris Anomalies

Condition	Description
Aniridia	Congenital absence or rudimentary iris; associated with Wilms' tumor (PAX6 gene)
Iris coloboma	Keyhole-shaped defect, usually inferonasal; failure of fetal fissure closure
Heterochromia	Difference in iris color between two eyes (binocular) or within one iris (sectoral)
Iris nevus	Flat, pigmented lesion on iris; benign but monitor for growth

Scleral Anomalies

Condition	Description
Staphyloma	Ectatic bulging of sclera lined by uveal tissue; common in high myopia
Episcleritis	Inflammation of episcleral tissue; sectoral or diffuse; self-limiting
Scleritis	Deeper inflammation of sclera; painful; associated with autoimmune disease

Palpation Test for IOP

Gross assessment of intraocular pressure; compares one eye to the other — NOT a replacement for tonometry.

Procedure:

1. Instruct patient to look down and close both eyes
 2. Place forefingers of each hand on the upper scleral portion under the orbital ridge
 3. Alternately apply gentle downward pressure with each finger
 4. Examiner closes own eyes or looks away to avoid visual bias on tactile judgment
 5. Record: hard / firm / soft
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6. Objective Refraction — Retinoscopy

Definition

Objective method of determining the refractive state without relying on patient responses. Also called: skiascopy, keratoscopy, shadow test, skiametry, pupilloscopy.

Principle (Foucault's method): The refractive state is determined by locating the eye's conjugate foci in space with a light source at a known point. The fundus reflex appears to be located in the plane of the patient's pupil.

Types of Retinoscopes

Type	Light Source	Advantage
Spot retinoscope	Round spot of light	Simple
Streak retinoscope	Linear filament (streak)	Preferred for astigmatism; can confirm motion by rotating streak

Important: The streak orientation is always **perpendicular to the meridian being scoped**.

Types of Retinoscopy

Type	Fixation Distance	Purpose
Static	6 metres (infinity)	Determines refractive state at distance
Dynamic	Near (40 cm or working distance)	Determines accommodative lag at near

Retinoscopic Reflexes

Reflex	Movement	Meaning	Action
With-motion	Reflex moves in SAME direction as streak	Focal point behind examiner (hyperopia / low myopia)	Add PLUS lens
Against-motion	Reflex moves in OPPOSITE direction	Focal point between patient and examiner (myopia > working distance)	Add MINUS lens
Neutral	Pupil lights up completely; no movement	Conjugate point = examiner's eye plane	NEUTRALIZED — subtract working distance

Strategy: Always convert against-motion to with-motion first, then approach neutrality from the with-motion direction.

Working Distance

Working Distance	Working Distance Lens to Subtract
50 cm	+2.00 D
67 cm	+1.50 D
1 metre	+1.00 D

Net finding = Gross finding - Working Distance Lens

Example: Gross = +2.00 D, WD = 50 cm (+2.00 D) → Net = **Plano**

Procedure: Static Retinoscopy — Spherical Method

1. Patient fixates a 6-metre target with non-tested eye
2. Examiner works at chosen working distance (usually 50 cm)
3. Shine streak across pupil; observe reflex motion
4. If against-motion: add minus sphere until with-motion
5. Add plus in +0.25 D steps until pupil lights uniformly (neutral reflex)
6. Record gross sphere power at neutrality
7. Repeat across perpendicular meridian if astigmatism suspected
8. Compute net finding by subtracting WD lens from gross
9. Measure VA with net retinoscopy finding in place

Sphero-Cylinder Method

1. Neutralize the **least myopic (most hyperopic) meridian first** using sphere
2. Neutralize the second meridian using minus cylinder added to the sphere
3. The streak is oriented perpendicular to the meridian being scoped

Worked example:

- At 180°: with-motion neutralized with +3.00 D
- At 90°: against-motion → add -0.75 cyl x 90 until neutral

- Gross: +3.00 sph / -0.75 cyl x 90
 - Net (WD = 50 cm, -2.00 D): **+1.00 sph / -0.75 cyl x 90**
-

Astigmatism — Special Phenomena (Streak Retinoscopy)

Phenomenon	Description
Break	Reflex not continuous with the streak on the patient's face
Thickness	Reflex thickness varies as streak rotates 360°
Skew	Reflex moves in a different direction than the streak (not aligned with principal meridian)
Scissors	Reflex appears to split in opposite directions — indicates irregular astigmatism

No skew phenomenon occurs with purely spherical refractive error.

Accuracy Errors in Retinoscopy

- Incorrect working distance (most common)
 - Scoping off the patient's visual axis
 - Failure of patient to fixate the distance target (accommodation interferes)
 - Failure to obtain a reversal (confirmation of neutrality)
 - Failure to locate the principal meridians correctly
 - Missing scissors motion (indicates irregular astigmatism)
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Dynamic Retinoscopy Methods

Used at near, typically at 40 cm (unless otherwise stated). **No working distance correction applied.**

MEM (Monocular Estimate Method) — Harold Haynes

Measures the accommodative lag at near working distance. Pre-cut plus lenses (in 0.25 D increments) are

briefly held before the eye; the power that eliminates the reflex movement indicates the lag.

- Normal lag: **+0.50 D to +0.75 D** (the eye under-accommodates slightly at near)

Nott Method

Target held at working distance; examiner moves back or forward until neutral reflex. No lens added.

OEP Method — Skeffington (Test #5)

Patient fixates a 20/40 letter or larger at working distance (typically 40 cm). Lag and lead of accommodation are noted with the habitual correction in place.

Sheard's Method

Examiner notes the retinoscopic reflex at two different working distances.

Tait's Method

Working distance: 13 inches or 16 inches (33 cm).

- Formula: **PEX = En - Ed**; **AC = TC - PEX**
- TC at 13" = 18Δ; TC at 16" = 15Δ

OEP Dynamic Retinoscopy Net Calculations

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Net #4 = Gross #4 - Working Distance (WD in dioptries)
Net #5 = Gross #5 - Lag
          (Modified if AA < 5.00D: Modified Lag = Lag × AA/5)
Net #6 = Gross #6 - Lag
Reading Add = Net #5 - Net #4
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7. Trial Case Accessories

Specialized devices for specific diagnostic purposes.

Pinhole Disc

1 or 2 mm hole in an opaque disc.

Purposes:

- Distinguish causes of poor vision (refractive vs. pathologic)
- Check the final correction
- Estimate nature and amount of refractive error

Clinical rule: Pinhole should IMPROVE acuity if the problem is refractive. No improvement or worsening = pathology.

Stenopaic Slit

0.5–0.75 mm wide aperture, up to 15 mm long. Limits light admission to approximately one meridian.

Use: Determine the amount and axis of the cylindrical component.

Scheiner's Disc

Two small pinholes spaced to fall within the average-sized pupil. A single point of light appears single only if rays focus on the retina; if off the retina, the point appears doubled.

Cobalt Disc

Cobalt glass absorbing the central spectrum and transmitting red and blue extremes. A white light at 1.4 m creates equal blur circles in emmetropia.

Refractive State	Cobalt Disc Appearance
Emmetropia	Purple circle with light blue fringes
Hyperopia	Blue circle with light red border
Simple hyperopic astigmatism	Purple ellipse with red extremities

Compound hyperopic astigmatism	Blue ellipse with red extremities
Myopia	Red circle with light blue border
Simple myopic astigmatism	Purple ellipse with blue extremities
Compound myopic astigmatism	Blue ellipse with blue extremities
Mixed astigmatism	Purple/red in one meridian, blue in the other
Irregular astigmatism	Irregularly shaped oval of irregular coloring
Too much (+) Rx	Reddish dot with purple fringe
Less (+) or too much (-) Rx	Bluish central dot with reddish border
Correct Rx	Purple area, no red dot in center

Red Lens

Interrupts fusion; used to detect diplopia or check for malingering. Placed before one eye with a red/green chart to test each eye independently.

8. Subjective Refraction

Technique of comparing lenses using changes in vision to arrive at the dioptric lens combination that results in maximum visual acuity. Uses the net retinoscopy finding as the starting point.

4 Basic Steps

1. Find the best sphere (fogging + defogging)
2. Determine the cylindrical component (axis and power)
3. Sphere refinement (Duochrome / bichrome test)
4. Binocular balancing

Methods of Relaxing Accommodation

Method	Description
Fogging	Most common non-cycloplegic method; add plus to blur, relaxing accommodation
Cycloplegia	Pharmacologic paralysis of ciliary muscle; dilated pupil
Cyclodamia	Non-cycloplegic fogging technique

Step 1: Fogging Technique

Add enough plus to reduce VA to approximately **20/60 to 20/80**, relaxing accommodation. The patient sees blurred but not completely dark.

Procedure:

1. Occlude left eye; present Snellen chart
2. Ask patient to fixate the largest letter (E)
3. Add plus in +0.25 D steps until VA reduces to ~20/60 to 20/80
4. This is the fog — accommodation is now relaxed
5. Begin defogging

Step 2: Defogging — Best Sphere

Reduce the fogging lens in -0.25 D steps (decrease plus for hyperopia; increase minus for myopia) until the patient achieves best VA without going past maximum plus / minimum minus. This is the MPMVA — maximum plus / minimum minus for best visual acuity.

Step 3a: Cylinder Axis — Two Methods

Method	When to Use	How It Works
Rotation Method	Cylinder power above 0.75 D	Rotate axis clockwise then counterclockwise until blur; average of two blur positions = refined axis

Crossed Cylinder Method	Cylinder power below 0.75 D	Handle of cross-cylinder placed parallel to tentative axis; flip; rotate BOTH cross-cylinder and axis toward RED DOT until equality
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Step 3b: Cylinder Power — Cross-Cylinder Method

The red dot of the cross-cylinder is placed parallel to the refined axis.

Patient Prefers	Meaning	Action
Position with RED dot parallel to axis	Cylinder is too WEAK	INCREASE minus cylinder power
Position with WHITE dot parallel to axis	Cylinder is too STRONG	DECREASE minus cylinder power
Both positions equal	Cylinder correct	No change

Memory: **Red dot = add cylinder** (red = stop = needs MORE). **White dot = reduce cylinder**.

Cylinder power interpretation using the clock-face target (Maddox clock test):

Finding	Interpretation
Original black line stays black after adding minus cyl	Cylinder correction too WEAK — add more
Original blurred line becomes blackest with minus cyl	Cylinder correction too STRONG — reduce
Blurred line does NOT darken with adding minus	IMPROPER FOG — re-fog and restart
Increasing minus produces no reversal; original black line returns	IRREGULAR astigmatism — use weakest cylinder that best equalizes

Step 3c: Sphere Refinement — Duochrome / Bichrome Test

Based on chromatic aberration: the eye focuses different wavelengths at different depths.

- **Green** (short wavelength) focuses **in front of** the retina
- **Red** (long wavelength) focuses **behind** the retina
- **Yellow** focuses exactly **on** the retina in emmetropia

Patient Reports	Hyperope Action	Myope Action
GREEN side clearer	INCREASE plus (under-corrected)	DECREASE minus (over-corrected)
RED side clearer	DECREASE plus (over-corrected)	INCREASE minus (under-corrected)
Both equal	Endpoint (correct)	Endpoint (correct)

Golden rule: If no point of equality is reached, ALWAYS leave **green clearer** (= more plus / less minus). Leaving green clearer protects against over-minusing and latent hyperopia.

Step 4: Binocular Balancing

Purpose: balance the state of accommodation between the two eyes after monocular refraction. Only done when both eyes have similar VA (within 1 line of each other).

Technique	How It Works	Best Use
Turville Infinity Balance (TIB)	Chart at 6 m reflected in a mirror; metal septum bisects so OD sees L, OS sees F; peripheral fusion maintained	Also detects heterophoria simultaneously
Prism Dissociation	3Δ BD on OD, 3Δ BU on OS; two vertically displaced targets	Most controlled method
Alternate Occlusion	Cover each eye alternately; compare clarity	Simplest but least reliable

Bichrome Method	Red/green filter for binocular equalization	Quick check
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Prism Dissociation — Procedure:

1. Add +0.50 to +0.75 D to sphere of BOTH eyes
2. Target: 20/30 row (one line below best monocular VA)
3. 3Δ BD on OD; 3Δ BU on OS
4. Ask: are both targets equally clear?
5. If unequal: add +0.25 D to the BETTER eye in steps until equal
6. If +0.25 D reverses preference: allow dominant eye to retain better vision
7. Once equal: reduce fog to best VA

TIB Heterophoria Indicators:

Pattern Seen	Phoria
L F (aligned)	Orthophoria
L F (F displaced further right)	Esophoria (uncrossed diplopia)
F L (F to the left of L)	Exophoria (crossed diplopia)
L displaced up from F	Right hypophoria / Left hyperphoria
L displaced down from F	Left hypophoria / Right hyperphoria
L tilted inward at top	Right incyclophoria
L tilted outward at top	Right excyclophoria

9. Slit-Lamp Biomicroscopy

Definition and Clinical Use

Clinical procedure used to examine the lids, anterior segment, and posterior segment of the eye. Called a "slit lamp" because the illumination system produces a tall, narrow slit of light — an optic section.

Clinical uses:

- Diagnostic and therapeutic visualization with stereoscopic view
- Contact lens fitting — evaluating fitting relationships with cornea, conjunctiva, and lids
- Foreign body removal from cornea and conjunctiva
- Measuring corneal thickness, anterior chamber depth, and IOP

Two Instrument Types

Type	Light Source Location
Zeiss slit lamp	Base of the instrument
Haag Streit	Top of the instrument

Memory: **Zeiss = Base. Haag Streit = Top.** (Very frequently swapped in MCQs.)

Basic Components

1. Illumination Arm — houses the illumination system (0–90° from straight ahead).

- Slit control: adjusts slit width and height/length
- Click stop: when engaged, slit beam focus coincides with viewing system focus
- Filters: cobalt blue, red-free (green), neutral density, diffusing

Filter	Appearance	Clinical Use
Cobalt blue	Blue light	Fluorescein staining; contact lens fitting
Red-free (green)	Red objects appear darker	Blood vessels, limbal arcades, neovascularization
Neutral density	Reduces intensity	General examination
Diffusing	Wide diffuse beam	Low-mag overview of eye and adnexa

2. Microscope Arm — houses the viewing system (normally straight ahead).

- Oculars: adjustable for examiner's refractive error and PD
- Magnification lever/changer: stepwise or continuous (zoom) system

3. Slit-Lamp Position Control

- Joystick / Elevation knob: controls horizontal movement and height

Six Illumination Techniques — Mnemonic: DR IS F

(Diffuse, Retro, Indirect, Specular, Filtered — plus Direct as the core technique)

Technique	Setup	What to Observe
Diffuse	30–45°, wide open slit, diffusing filter, low-medium magnification	General view of anterior eye and palpebral conjunctiva; contact lens centration and movement
Direct (Optic Section)	30–45°, very narrow focused slit	Corneal curvature variation, corneal thickness, depth of foreign body
Direct (Parallelepiped)	30–45°, wider focused slit (~3 mm)	Corneal stroma, epithelial breakdown, lens fit, lens surface, endothelium
Indirect	30–45°, moderate beam, slit offset beside area of interest	Corneal microcysts, vacuoles, epithelial anomalies
Retro-illumination	Light reflected off iris or fundus; microscope focused on cornea	Corneal neovascularization, edema, microcysts, vacuoles, infiltrates; contact lens surface deposits
Specular Reflection	60° angle , parallelepiped, move illumination arm until bright reflex appears (angle in = angle out)	Individual endothelial cells (polymegethism, guttata), tear layer stability and lipid layer
Filtered (Cobalt Blue)	Cobalt blue + Wratten filter (#12, yellow)	Corneal staining patterns; RGP contact lens fitting

Important: Specular reflection is the ONLY illumination technique that requires a specific 60° angle. Van Herick also uses 60° but for a different purpose.

Slit-Lamp Examination Routine

Examination order: **Lids → Lashes → Conjunctiva → Limbus → Cornea → Tears → Aqueous → Iris → Crystalline Lens**

1. Set up slit lamp: focus eyepieces, adjust headrest, position fixation target
 2. Decrease room illumination
 3. Start with DIFFUSE illumination at LOW magnification
 4. Systematically examine all anterior structures
 5. Switch to other illumination techniques as needed
 6. Van Herick at temporal limbus: 60° angle, optic section, assess anterior chamber depth
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Van Herick Technique

Estimates depth of the anterior chamber to screen for angle closure risk.

Setup: Slit lamp at **60°**, narrow the beam to an optic section, patient looks straight ahead, focus beam sharply on cornea at the temporal limbus. Compare the width of the dark "shadow" on the iris (the optically empty aqueous) to the width of the corneal optic section.

Grade	Shadow : Cornea Ratio	Clinical Meaning
Grade 4	$\geq 1:1$	Open anterior chamber angle — safe
Grade 3	Between 1:2 and 1:4	Open angle — monitor
Grade 2	Exactly 1:4	Narrow angle — caution
Grade 1	$< 1:4$	Susceptible to angle closure — high risk
Grade 0	No shadow; cornea "sits" on iris	Angle is CLOSED — active angle-closure attack

Memory: Smaller grade = more danger. Grade 0 = attack. Grade 1 = risk only (not attack).

The shadow represents the optically empty aqueous in the anterior chamber, NOT the cornea itself.

objective examination of the interior of the eye with an ophthalmoscope

10. Ophthalmoscopy

Direct vs Indirect Ophthalmoscopy

Feature	Direct	Indirect
Image type	Virtual and ERECT	Real and INVERTED
Magnification	~15×	2–5× (5× with 12D; less with higher power)
Field of view	10° (2 disc diameters)	37° with 20D (8 disc diameters)
Area visible	Not beyond equator	Between equator and ora serrata visible
Maximum resolving power	70 μm	20 μm
Working distance	1–2 cm (very close)	Arm's length
Stereopsis	None (monocular)	Excellent (binocular)
Illumination	Poorer	Better
Fundus image	Fixed, limited	Wide, peripheral access

Condensing lens for indirect: Higher power = lower magnification + larger field of view. Most commonly used: **+14D** and **+20D**.

Procedure: Direct Ophthalmoscopy

1. Reduce room illumination
2. Set lens wheel to 0 D (compensate for examiner's refractive error)
3. For right eye: examiner uses RIGHT eye, holds in RIGHT hand; approach from temporal side at ~15° from line of sight
4. Begin at arm's length; look for red reflex
5. Move in to 1–2 cm; focus on optic disc
6. Evaluate: disc margin, color, cup-to-disc ratio
7. Follow four main vascular arcades
8. Examine macula last (bright light is uncomfortable); ask patient to look directly at the light

Opacity Localization by Parallax (Direct Ophthalmoscopy)

Move the ophthalmoscope from side to side:

Opacity Movement (vs. ophthalmoscope)	Location
Opposite direction	Anterior (cornea, anterior chamber, anterior lens)
Little or no movement	Near the nuclear lens
Same direction	Posterior lens or vitreous
Vitreous: floats back to original position	Vitreous

When the PATIENT moves the eye: anterior opacities move in the SAME direction; posterior opacities move in the OPPOSITE direction.

Normal Fundus Appearance

Structure	Normal Appearance
Optic disc	Round to oval; 1.5–1.7 mm diameter; red-yellow, nasal side more pronounced; sharp margins
Vessels emerge from	Within the physiologic cup/excavation
Retinal periphery	Lighter than central area; occasional pigmentation irregularities
Fovea	Appears darker than surrounding retina
Macula	Yellow color visible only in red-free illumination
Arteries	Light red, straight, white reflex stripes
Veins	Dark red, tortuous, pulsating
Artery : Vein caliber	2:3 (veins larger)
Choroidal vessels	White/pink/yellow-white; very tortuous; numerous anastomoses

Ophthalmoscope Accessories

Light apertures:

- Micro-spot (small): undilated, well-lit room
- Medium aperture: dark room, non-dilated pupil (most commonly used)
- Large aperture: dilated pupil, general examination
- Half-light: partially obstructed pupil (e.g., cataract)
- Slit beam: examines contour abnormalities of cornea, lens, retina
- Grid: rough approximation of relative distance between retinal lesions

Ophthalmoscope filters:

- Cobalt blue (violet): fluorescein staining, contact lens evaluation
 - Red-free (yellowish-green): enhances contrast with red; best for blood vessels
 - Yellow filter: decreases light intensity; provides comfort for patient's retina
-

Comprehensive Glaucoma Examination — "TOP GP"

Test	Measures	Instrument
Tonometry	Intraocular pressure (normal: 12–22 mmHg)	Contact / non-contact tonometer
Ophthalmoscopy	Shape and color of optic nerve	Dilated fundus exam
Perimetry	Complete visual field map	Automated perimeter
Gonioscopy	Angle where iris meets cornea (open vs. closed)	Gonioscopic contact lens
Pachymetry	Central corneal thickness (CCT)	Pachymeter

Gonioscopy: A hand-held contact lens with a mirror placed on the numbed eye. Shows whether the angle is open (open-angle / chronic glaucoma) or closed (angle-closure / acute glaucoma). **Required to differentiate open-angle from angle-closure — cannot determine this with tonometry alone.**

Pachymetry: Thin CCT is a strong predictor of glaucoma development in patients with borderline IOP.

Thin CCT can cause **underestimation** of IOP; thick CCT can **falsely inflate** tonometry readings.

Normal IOP: 12–22 mmHg. Most glaucoma cases exceed 20 mmHg, but normal tension glaucoma exists within the normal range.

Imaging for glaucoma:

- Scanning Laser Tomography (HRT3) — evaluates optic nerve head
 - Laser Polarimetry (GDX) — measures retinal nerve fiber layer thickness
 - Optical Coherence Tomography (OCT) — both time-domain and spectral-domain
-

11. Keratometry

Definition

Clinical technique used to measure the curvature of the anterior surface of the cornea. Determines power, toricity, and integrity of the cornea. Uses the principle of a convex mirror and a doubling principle.

Information Provided

- Radius of curvature of the anterior corneal surface
- Direction of the eye's principal meridians (with-the-rule or against-the-rule astigmatism)
- Axis and degree of corneal astigmatism
- Any corneal distortion

Optical Principle

The relationship between the size of an object and the size of its image reflected from a convex mirror. The doubling principle allows the examiner to measure the radius of the cornea without the image moving during the measurement.

Astigmatism Direction

- **With-the-rule (WTR):** Weakest corneal meridian at or near 180° (vertical meridian has more power). The minus axis of the correcting cylinder is at 180°.
- **Against-the-rule (ATR):** Weakest corneal meridian at or near 90°. The minus axis of the correcting cylinder is at 90°.

Sample recording:

R: 42.00 at 180 / 43.00 at 90 → Corneal astigmatism: -1.00 cyl x 180
L: 42.25 at 170 / 43.50 at 80 → Corneal astigmatism: -1.25 cyl x 170

Javal's Rule

Estimates total refractive astigmatism from corneal astigmatism:

$$A_t = p(A_c) + k$$

Where:

- **A_t** = total (refractive) astigmatism
- **A_c** = corneal astigmatism
- **p** = 1.25 (the cornea contributes approximately 1.25× its measured power to the total)
- **k** = -0.50 cyl x 90 (physiologic ATR astigmatism from the posterior corneal surface)

Simplified: $A_t = 1.25(A_c) + (-0.50 \text{ cyl} \times 90)$

Sources of Keratometry Error

- Improper calibration
- Faulty patient positioning
- Lack of proper fixation by patient
- Reduced examiner visual acuity
- Accommodative fluctuation by examiner
- Localized corneal distortions, excessive tear fluid, abnormal lid position
- Improper focusing of the corneal image

12. Phorometric Test — OEP 21 Steps

Two Visual Patterns

Pattern	Definition	Buffer Area
Accommodative Pattern	Ability to see clearly at varying distances	Hyperopia buffers accommodation
Convergence Pattern	Mechanism for maintaining single vision	Exophoria buffers convergence

The phorometric test investigates the relationship between these two patterns using probe bodies (lenses and prisms).

Probe Bodies — 4 Basic Changes

Probe Body	Pattern Affected	Induced Change	Perceived Size
Convex lens (+)	Inhibits accommodation	Object appears to move AWAY	Target appears BIGGER
Concave lens (-)	Stimulates accommodation	Object appears to move CLOSER	Target appears SMALLER
Base-In prism	Inhibits convergence	Object appears to move AWAY	Target appears BIGGER
Base-Out prism	Stimulates convergence	Object appears to move CLOSER	Target appears SMALLER

Memory shortcut: Plus lens and BI prism both **INHIBIT** (object moves away, bigger). Minus lens and BO prism both **STIMULATE** (object moves closer, smaller).

Phoria Measurement Rules

- **Lateral phoria:** use a **VERTICAL dissociating prism**
- **Vertical phoria:** use a **HORIZONTAL dissociating prism**
- Phorias measured at 20 ft (6 m) and 16 inches (40 cm)
- At 16 inches: stimulus to accommodation = 2.50 D; stimulus to convergence = 15 prism dioptres (if PD ~64 mm)

Von Graefe method: Dissociating prism placed in front of one eye; measuring prism in front of the other eye.

Standard prism setup (for tests #3, 8, 12, 13A, 13B, 15A, 15B, 18): OD: 12–15Δ BI; OS: 6Δ BU

The 21 OEP Steps – Complete Summary

Group 1: Lateral Phorias (Far and Near)

Test	Distance	Lenses In Place	Prisms	Endpoint	Significance
#3	20 ft / 6 m	Habitual correction	OD:12–15BI; OS:6BU	Vertical alignment	Lateral muscular imbalance at FAR (habitual)
#8	20 ft / 6 m	Test #7 lenses	OD:12–15BI; OS:6BU	Vertical alignment	Lateral imbalance at far as influenced by #7
13A	16" / 40 cm	Habitual near correction	OD:12–15BI; OS:6BU	Vertical alignment	Lateral muscular imbalance at NEAR (habitual)
13B	16" / 40 cm	Test #7 lenses	OD:12–15BI; OS:6BU	Vertical alignment	Lateral imbalance at near as influenced by #7

Group 2: Vergences at Far

Test	Name	Prism	Endpoint	Significance
#9	True Adduction / Advergence	OU ready BO; +0.25 aux sphere	Blur (same as +0.25)	Limit of available fusional convergence in reserve

#10	Positive Fusional Reserve at Far	OU ready BO	Break / Recovery	Breakpoint = reserve of total positive convergence at far
#11	Negative Fusional Reserve at Far (Abduction)	OU ready BI	Break / Recovery	Negative fusional convergence (divergence) in reserve

Group 3: Vertical Phoria and Duction

Test	Distance	Phoria Endpoint	Duction Endpoint	Significance
#12	20 ft / 6 m	Horizontal alignment	Break / Recovery	Vertical imbalance at far; reserve used to compensate
#18	16" / 40 cm	Horizontal alignment	Break / Recovery	Vertical imbalance at near; reserve used to compensate

Group 4: Accommodation at Near

Test	Name	Condition	Auxiliary	Endpoint	Significance
14A	Dissociated Cross-Cylinder	Test #7 lenses + OD:3BD, OS:3BU	Cross-cyl -0.50D, red dot at 90°	Equality (highest plus)	Lag of accommodation — DISSOCIATED condition
14B	Binocular Cross-Cylinder	Gross 14A (no dissociating prisms)	Cross-cyl -0.50D, red dot at 90°	Equality (highest plus)	Lag of accommodation — FUSED condition

15A	Induced Phoria via Unfused CC	Gross 14A lenses	None	Vertical alignment	Influence of 14A lenses on lateral imbalance at near
15B	Induced Phoria via Fused CC	Gross 14B lenses	None	Vertical alignment	Influence of 14B lenses on lateral imbalance at near

Memory: **A = Apart (dissociated). B = Both together (fused).**

Group 5: Relative Convergence at Near

Test	Name	Prism	Endpoint	Significance
16A	Positive Relative Convergence (PRC)	OU ready BO	Blur out	Amount of positive fusional innervation in reserve
16B	Positive Fusional Reserve (PFR)	OU ready BO	Break / Recovery	Total reserve of positive fusional convergence at near
17A	Negative Relative Convergence (NRC)	OU ready BI	Blur out	Amount of convergence relaxable at that fixation distance
17B	Negative Fusional Reserve (NFR)	OU ready BI	Break / Recovery	Amount of accommodative convergence that can be inhibited

Memory: **A tests (16A, 17A) end at BLUR. B tests (16B, 17B) end at BREAK/RECOVERY.**

Group 6: Accommodation Tests

Test	Name	Endpoint	Significance
#19	Amplitude of Accommodation	Presbyope: first readable; Non-presbyope: print difficult	Maximum accommodation amount
#20	Positive Relative Accommodation (PRA)	Blur out (adding minus lenses)	Maximum increase in accommodation while convergence is fixed
#21	Negative Relative Accommodation (NRA)	Blur out (adding plus lenses)	Maximum relaxation of accommodation while convergence is fixed

Key Cross-Session Links

- **Van Herick Grade 0** (angle closed) links to **Gonioscopy** — confirms angle-closure glaucoma
- **Net retinoscopy finding** = starting point for subjective refraction (not gross finding)
- **Duochrome endpoint** depends on full accommodation relaxation — if fogging was inadequate, re-fog
- **TIB** simultaneously balances accommodation AND detects heterophoria
- **Thin CCT + borderline IOP** = increased glaucoma risk; thin CCT causes underestimation of IOP
- **Tests 14A and 14B** use the minus cylinder from objective refraction as the starting lens power

13. High-Yield Cheat Sheet and Board Traps

Quick Numbers Reference

Value	What It Is
20 fc	Standard chart illumination
80%	Snellen chart contrast
1/5	Snellen letter stroke-width ratio

1/2	Snellen letter spacing and row separation ratio
+0.50 to +0.75 D	Normal MEM accommodative lag
5 cm ± 2.5	NPC break point (accommodative target)
7 cm ± 4	NPC recovery (accommodative target)
12–22 mmHg	Normal IOP range
60°	Specular reflection AND Van Herick technique angles
15× / 10°	Direct ophthalmoscopy: magnification / field
2–5× / 37°	Indirect ophthalmoscopy: magnification / field
3:2	Vein-to-artery diameter ratio in normal fundus
1.5–1.7 mm	Normal optic disc diameter
64/61 mm	Typical adult distance/near PD
33 cm	Amsler grid working distance
15 – 0.25(age)	Hofstetter minimum amplitude of accommodation
4 or fewer wrong	Normal Ishihara score
5 or more wrong	Defective red-green color vision (Ishihara)
2 crossings ≥ 4 caps	Farnsworth D-15 failure criterion
16 caps	Farnsworth D-15 total caps
85 caps	FM-100 Hue total caps
10Δ per mm	Hirschberg reflex displacement equivalence

Board Traps — The Most Tested Errors

Case History

- HIGH blood sugar = MORE MYOPIC (not more hyperopic). The lens swells, shifting the focal point anterior.
- Asthenopia = eyestrain. Amblyopia = reduced VA. Not interchangeable.
- Not all headaches are refractive. Location and triggers matter. Occipital headaches may be

convergence insufficiency or systemic, not astigmatism.

Visual Acuity

- Snellen acuity is minimum SEPARABLE (resolution), not minimum visible.
- Scoring stop rule: more than HALF the letters on a row missed — not the first row with any missed letter.
- Easiest letter = L. Hardest letter = B.
- Pinhole that does NOT improve VA suggests pathology, not refractive error.

Color Vision

- Ishihara tests ONLY red-green deficiencies. Blue (tritan) defects require HRR.
- Plate 1 of Ishihara is a DEMONSTRATION plate — failure may indicate very low VA or malingering, not color deficiency.

Retinoscopy

- Scissors motion = IRREGULAR astigmatism. Cannot be corrected with a standard cylinder.
- Streak orientation is PERPENDICULAR to the meridian being tested (not parallel).
- Dynamic retinoscopy does NOT subtract a working distance lens.
- Against-motion does not always mean high myopia — the patient may be over-minused or highly accommodating.

Subjective Refraction

- RED dot on cross-cylinder = INCREASE cylinder. WHITE = DECREASE. Commonly reversed in MCQs.
- Blurred line does NOT darken with adding minus cylinder = IMPROPER FOG, not irregular astigmatism. Re-fog first.
- Rotation method for axis: ABOVE 0.75 D. Cross-cylinder method: BELOW 0.75 D.
- If duochrome equality cannot be reached, leave GREEN clearer — never red. Red clearer = over-minused.
- Dominant eye is allowed to retain better vision if +0.25 D reversal occurs in binocular balancing.
- Alternate occlusion is the simplest but LEAST reliable balancing method.

Slit-Lamp

- Zeiss = light at BASE. Haag Streit = light at TOP. Frequently swapped in MCQs.
- Specular reflection requires EXACTLY 60°. Van Herick also uses 60° but for a different purpose.
- Direct (optic section) = very narrow slit. Direct (parallelepiped) = ~3 mm wide. Not the same setup.
- Retro-illumination: light reflected off IRIS or FUNDUS; microscope focuses on CORNEA, not iris.
- Click stop: when engaged, slit beam and viewing system are co-focused.

Van Herick

- Grade 1 = susceptible to closure, NOT an active attack. Grade 0 = active angle-closure attack.
- The shadow represents optically empty aqueous, NOT the cornea.

Ophthalmoscopy

- Direct gives a VIRTUAL, ERECT image. Indirect gives a REAL, INVERTED image. Most commonly swapped MCQ answers.
- Indirect sees equator to ora serrata. Direct does NOT.
- IOP within 12–22 mmHg does NOT rule out glaucoma. Normal tension glaucoma exists.
- Thin CCT = underestimated IOP. Thick CCT = falsely elevated IOP.
- Gonioscopy is required to distinguish open-angle from angle-closure glaucoma. Tonometry alone cannot do this.
- Higher condensing lens power in indirect ophthalmoscopy = LOWER magnification + LARGER field.

Phorometric Tests (21 Steps)

- 14A = DISSOCIATED cross-cylinder (prisms separate the images). 14B = FUSED (no dissociating prisms).
- PRC/NRC (16A/ 17A) endpoints = BLUR. PFR/NFR (16B/ 17B) endpoints = BREAK/RECOVERY.
- Buffer areas: Hyperopia buffers accommodation. Exophoria buffers convergence. Different patterns.
- Test #9 requires an auxiliary +0.25 D lens. Tests #10 and #11 do not.
- Vertical phoria uses a HORIZONTAL dissociating prism. Lateral phoria uses a VERTICAL dissociating prism.

*Sources: Dr. Ferrer Primary Care Optometry Complete Notes (MCU 2024, 104 pp.) · PCO Board Reviewer
10 Sessions / 20 Hours · PrimaryCareOptometry Program Assessment Reviewer (Dr. Wyrlo, Pages 5–103) ·
Personal consolidation for 2026 OLE Board Review*